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A low Cost Smart Insole for Diabetic Foot Prevention

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Abstract— **The foot pressure measurement is important in gait and posture research for diagnosis of pressure of the feet for the athletes, back injuries, adaptation of insoles for orthopaedic applications and prevention of diabetic foot ulceration. There are several methods to measure the distribution of foot pressure. However, traditional systems have the disadvantage of being no flexible and thus influencing the measurement. In this work we present a low cost smart insole for diagnosis and prevention of plantar perforating disease in diabetic patients. The system consists of an instrumented insole that contains innovative sensors witch allow the quantification of plantar pressure at critical points that have a high risk of ulceration, it also contains a temperature / humidity sensors. A high rate of temperature and humidity allows faster formation of the ulcer, especially in the high friction points. All sensors for both feet communicate wirelessly and in real time with a mobile application through Bluetooth.**

Keywords— insole, plantar, pressure, ulceration, gangrene, diabetic foot, temperature, humidity.

I. Introduction

Diabetes is one of the challenges facing the health care sector today. One of the major diseases of diabetic patients is peripheral neuropathy where diabetic patients develop pressure ulcers in their feet [1]. It is estimated that 15% of diabetic patients develop foot ulcerations and more than 85% of foot amputations in cases of diabetes are due to foot ulcerations [2]. Most foot ulcers are the consequence of a trauma and followed by amputation [3]. Previous studies have suggested that effective care could reduce foot ulcer and amputation [4]. A bad plantar pressure distribution, an excessive humidity and a high temperature are factors depending of the quality of shoes that affect feet and may cause disease like ulcer. A hot and humid environment is factors for the development of virus and microbes. A high pressure in specific point of the insole increases the risk of these pathologies. There is a need to measure all these parameter simultaneously in a daily shoe for feet's disease prevention [5].

II. Architecture System

The aim of our system is to provide a flexible, low-cost solution with precise and real-time measurements. We used a Smartphone for data processing and visualization tasks for both feet, in order to be informed about the risk of ulcer formation during the patient's daily activities and warns in case of danger. For each foot, the system consists of an instrumented insole pressure distribution sensor, with three coupled humidity / temperature sensors (Fig.1). The insoles pressure distribution sensors are developed in our laboratory, which contains eight pressure sensors developed by ourselves based on Hall Effect sensors. The temperature and humidity sensors are located in a high risk area. For both feet, the communication is wirelessly with the mobile application through Bluetooth. The data of all sensors acquired by the mobile application are represented in an event list, in real time graphs and in SVG images.

Fig.1 A global architecture

The pressure sensors are connected to an eight-input multiplexer, the output of the multiplexer is connected to a conditioning circuit which contains an amplification stage followed by an active low-pass filtering, the output of the filter is connected to an Analog input in the microcontroller, and the temperature / humidity sensors communicate with the microcontroller with an I2C bus (Fig.2).

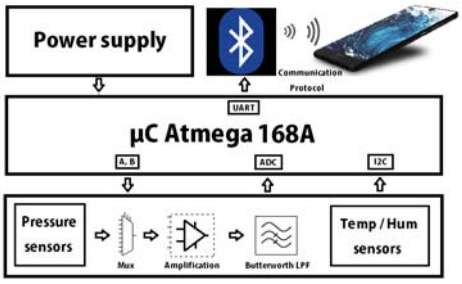

Fig. 2 A Hardware architecture

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III. Sensor insole

To have a good flexibility that protects the foot of the patient and allows having a precise measurement through the insole. This has been developed using silicone and contains eight pressure sensors that can be removed and placed in different insole according to the size of the foot (Fig.3).

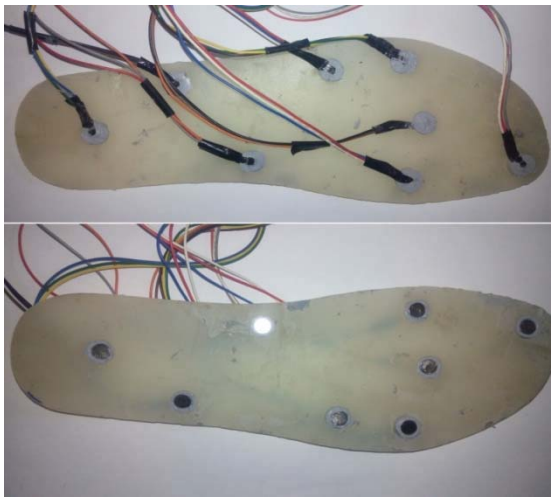

Fig. 3 Sensor insole

The transducer consists of a HALL sensor and a permanent magnet separated by a thin layer of 5 mm of elastic polymer material [6]. Under the effect of a compressive force the sensor approaches the magnet. Fig.4 illustrates the designed transducer based by UGN3503 sensor [7]. This is used to measure magnetic flux densities, linear and angular position, and rotational speed. [8]. Fig.5 illustrates the transducer element realized.

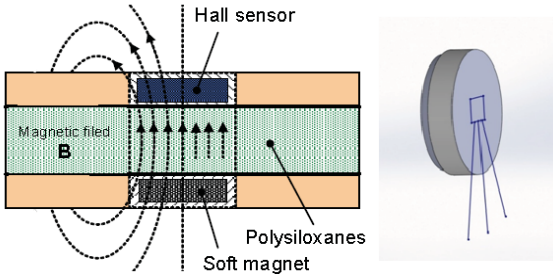

Fig. 4 Transducer design

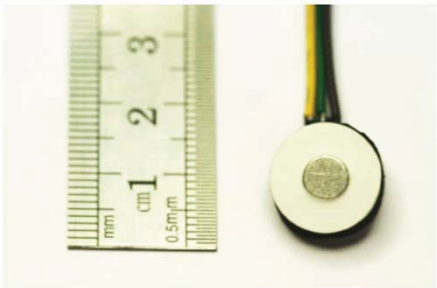

Fig. 5 Transducer realized

IV. Humidity and temperature sensors

A three coupled temperatures / humidity SHT1X sensors are placed in the shoe where the friction between the skin and the shoe is high. High temperatures with loss of sensation predispose to foot ulceration [9], [10].The frictions can cause skin damage and may end with injuries which increase the risk of ulcer. Studies show that a difference of 2.2°C between two risky areas may become an ulcer [11].

Temperature and humidity, termed as 'microclimate' between the skin and the supporting surface, are important extrinsic factors implicated in ulcer development, together with pressure and shear. This was confirmed in a mathematical study (Gefen 2011), which demonstrated that an increase in humidity levels around the skin, decreases skin tolerance to superficial pressure. The authors stated that an increase in RH by 25%, decreased skin tolerance by 24%, making it more vulnerable to pressure ulcer formation [12].

V. Hardwar Card

For each foot, all the sensors are connected to an acquisition card. The microcontroller used is ATMEGA168A with a clock frequency of 16MHz. The pressure sensors are connected to a multiplexer CD74HC4051, followed by an instrumentation amplifier AD620, with an amplification gain equal to 2, which is followed by a low pass filter, based on an operational amplifier OP07 with a cut-off frequency of 160 Hz and an amplification gain equal to 2. The sensors temperature / humidity are connected to the microcontroller by the I2C bus, a power supply stage containing a 5V LM7805 regulator, the + 5V voltage is reversed by ICL7660, to have a voltage of -5V, the power supply is provided by two 3.7V batteries. The embedded acquisition card in the shoe is presented in Fig.6.

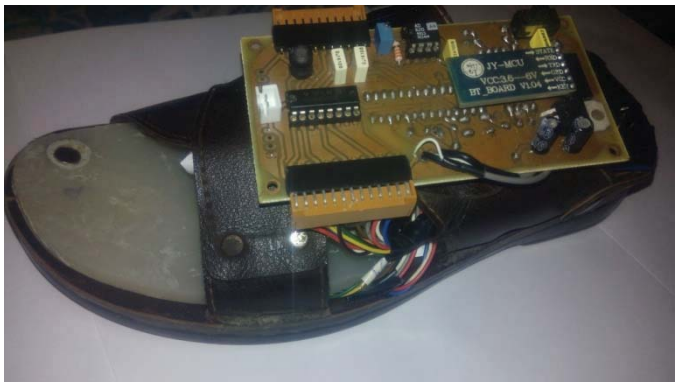

Fig. 6 Embedded card in the shoe

The communication between the acquisition cards and the mobile application is provided by a communication protocol, the format of the frame is represented in the following :

Start	Foot address	Sensor address	Data	Checksum	Stop
1 byte	1 byte	1 byte	2 bytes	1 byte	1 byte

Fig. 7 Frame format

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VI. Smart Mobile Application

The mobile application allows the diabetic patient to have an analysis of the plantar pressure in static and in dynamic daily activities, so the patient can be informed about the state of his foot and warns in case of danger. The application contains several windows, for authentication, to activate the Bluetooth of the phone, to search a different Bluetooth enabled devices, to connect and disconnect with devices already found. It also contains a menu allowing the user to access all the options of the application (Fig.8).

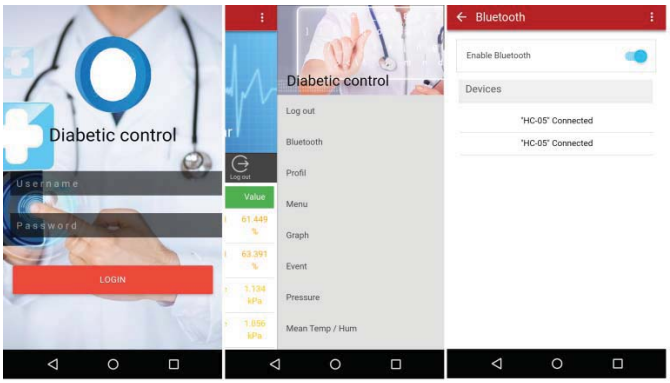

Fig. 8 Authentication, menu and Bluetooth windows

The application also contains a window that represents the patient's profile with an event list, contains real-time alerts to warn the patient about the degree of danger, and the alerts are processed for both feet by three thresholds for each sensor (Fig.9).

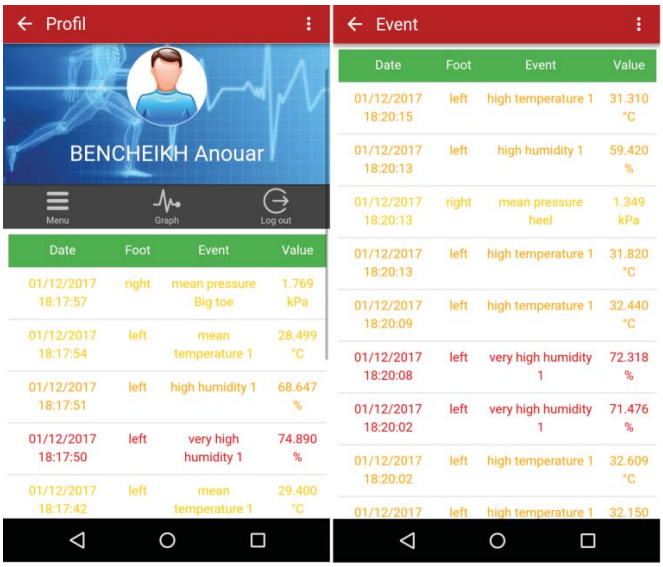

Fig. 9 Profile and event windows

The application also contains a menu that provides access to the event list and different SVG images. An SVG image represents the pressure applied in real-time on the different points of the foot, when you click on one of these points you can monitor the pressure in a real-time graph (Fig.10).

Fig. 10 Menu and real-time pressure measurement windows

For each foot, the temperature and humidity can be monitored in an SVG image that contains the average value for the three temperature and humidity sensors, in another window we can monitored in a real-time the value for each sensor, you can click on one of the three points to monitor temperature and humidity in real-time graphs (Fig.11).

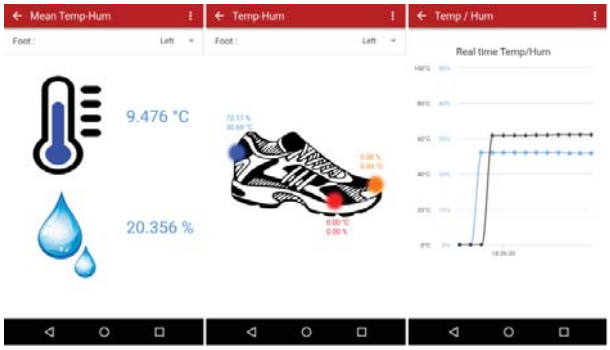

Fig. 11 Real-time temperature and Humidity measurement

To monitor the real-time variation of several graphs at the same time, we can configure up to an eight graphs. For each graphs, the foot, the type, and the name of the point must first be selected (Fig.12).

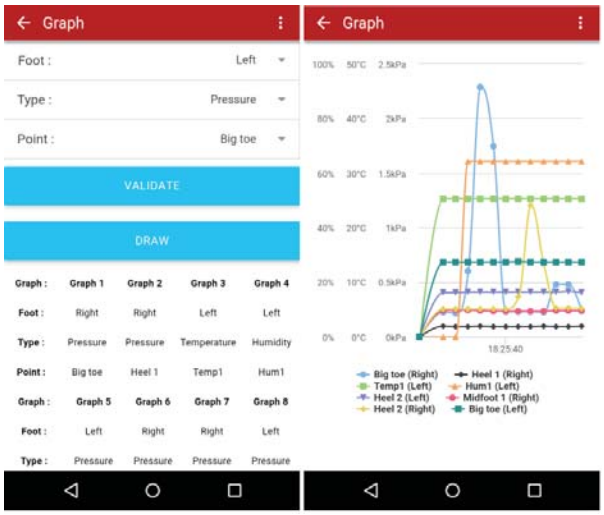

Fig. 12 Configuration and real-time measurements windows

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VII. Conclusion

An instrumented insole with specified sensors has been developed for monitoring foot pressure distribution. This instrument can be used as intelligent footwear in order to prevent ulcer formation of diabetics. Currently, tests are underway for variable pressure applied in the required range and the same for variable size of foot sole. The developed system is flexible and portable for field studies and also advantageous due to large memory size, dynamic recording of pressure and no fear of foot step modification.

Acknowledgment

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